

JEE MAIN REVISION MASTER: INVERSE TRIGONOMETRIC FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Inverse Trigonometric Functions — Core Principles & Governing Laws	31 PYQs	Core Principle
Moderate	Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	6 PYQs	Quantitative Analysis
Standard	Inverse Trigonometric Functions — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Inverse Trigonometric Functions — Core Principles & Governing Laws	$\cos(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{33}{65}), \quad \text{f(x)} = \sin^{-1}(\frac{x^2 - 3x + 2}{x^2 + 2x + 7}), \quad \text{a} = \sin^{-1}(\sin(5))$
Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	$\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2}), x \in (0,1), \quad 2 - 2x^2 = 1 + 2x^2, \quad \alpha = \sin^{-1}(\frac{2}{\sqrt{13}})$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Inverse Trigonometric Functions — Core Principles & Governing Laws (Very High Yield — 31 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Inverse Trigonometric Functions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Inverse Trigonometric Functions PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha = 3\sin^{-1}((6/11))$ and $\beta = 3\cos^{-1}((4/9))$, where inverse trigonometric functions take only the principal value..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the domain of the function $f(x) = \sin^{-1}(\frac{1}{x^2-2x-2})$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta =$..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "The number of solutions of $\tan^{-1}4x + \tan^{-1}6x = (\pi/6)$, where $-\frac{1}{\sqrt{6}} < x < \frac{1}{\sqrt{6}}$, is equal to :..." [Ans: N/A]

3.2 Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes (Moderate Yield — 6 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "If $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2})$, $x \in (0,1)$, then the value of x is:..." [Ans: A]
- [2026 Evening Shift] Q#4 (MCQ): "Considering the principal values of inverse trigonometric functions, the value of the expression $\tan(2\sin^{-1}(\frac{2}{\sqrt{13}})) - 2\cos^{-1}(\dots)$ " [Ans: N/A]
- [2025 Evening Shift] Q#4 (MCQ): "The value of $\cot^{-1}(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}) - \cot^{-1}(\frac{\sqrt{1+\tan^2(1/2)}+1}{\tan(1/2)})$ is equal to..." [Ans: N/A]

3.3 Inverse Trigonometric Functions — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha = 3\sin^{-1}((6/11))$ and $\beta = 3\cos^{-1}((4/9))$, where inverse trigonometric functions take only the principal value..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the domain of the function $f(x) = \sin^{-1}(\frac{1}{x^2-2x-2})$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta =$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Let $\alpha = 3\sin^{-1}((6/11))$ and $\beta = 3\cos^{-1}((4/9))$, where inverse trigonometric functions take only the principal values. Given below are two statements : Statement I : $\cos(\alpha + \beta) > 0$. Statement II : $\cos(\alpha) < 0$. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: $(1/2) < (6/11) < \frac{1}{\sqrt{2}} \Rightarrow \sin^{-1}((1/2)) < \sin^{-1}((6/11)) < \sin^{-1}(\frac{1}{\sqrt{2}}) \dots (i) \Rightarrow 3\sin^{-1}((1/2)) < 3\sin^{-1}((6/11)) < 3\sin^{-1}(\frac{1}{\sqrt{2}}) \Rightarrow (\pi/2) < \alpha < (3\pi/4) \Rightarrow \cos\alpha < 0 \dots (ii) (4/9) < (1/2) < \frac{1}{\sqrt{2}} \Rightarrow \cos^{-1}((4/9)) > (\pi/3) > (\pi/4) \Rightarrow 3\cos^{-1}((4/9)) > \pi > (3\pi/4) \Rightarrow \beta > \pi \dots (ii)$ Inequalities (i) and (ii) shows the mathematical logic for $\alpha + \beta > \frac{\pi}{2}$

Benchmark Problem 2: JEE Main 2026 (Online) 6th April Evening Shift (Q#2)

If $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2})$, $x \in (0,1)$, then the value of x is:

- (A) $(1/2)$ **[Correct]**
- (B) $(1/3)$
- (C) $(2/3)$
- (D) $(5/8)$

Correct Answer: A

Solution Analysis: Let us solve $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2})$, $x \in (0,1)$ step by step. Step 1: Simplify the left-hand side Let $\theta = \tan^{-1}(x\sqrt{2})$ Then $\tan\theta = x\sqrt{2}$ We know that if $\tan\theta = (\text{opposite} / \text{adjacent})$, then we can take opposite $= x\sqrt{2}$ adjacent $= 1$ So hypotenuse is $\sqrt{(1+(x\sqrt{2})^2)} = \sqrt{1+2x^2}$ Hence, $\sin\theta = \frac{x\sqrt{2}}{\sqrt{1+2x^2}}$ Therefore, $\sin(\tan^{-1}(x\sqrt{2})) = \frac{x\sqrt{2}}{\sqrt{1+2x^2}}$ Step 2: Simplify the right-hand side Let $\phi = \sin^{-1}\sqrt{1-x^2}$ Then $\sin\phi = \sqrt{1-x^2}$ S